

Module 4: Search Theory

PhD Macro Qualifying Exam Notes

Macro Qual Lecture Notes Builder

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Source Inventory

Source	What it contributes	How it is used in this module
GitHub source folder (link)	Contains the course search note <code>3_Search.pdf</code> and Problem Set 3 <code>PS3-6.pdf</code> .	Establishes the local source corpus for Module 4.
Natalia Kovrijnykh, <i>3 Search Theory</i> , ECN 713, Jan. 30, 2025	Basic McCall model, reservation wage proof, unemployment duration, comparative statics in b , β , and wage dispersion, alternative U formulation, layoffs, aggregate unemployment dynamics, endogenous search intensity.	Main backbone for Sections 2–5; figures are redrawn in TikZ and formulas are expanded for exam use.
Problem Set 3, ECN 713 Spring 2026	Four exam-style search problems: no-offer probabilities by aggregate state; stochastic unemployment benefits; two offers with recall; temporary jobs with state-dependent offer arrival.	Used for the solved templates in Section 12.
Ljungqvist and Sargent, <i>Recursive Macroeconomic Theory</i>	Search, unemployment, recursive methods, McCall search, waiting times, firing/layoffs, lake dynamics, and related applications.	Used as the conceptual reference for recursive formulation, value measurement, and stock-flow unemployment dynamics.
McCall (1970), “Economics of Information and Job Search”	Canonical reservation wage logic for intertemporal job search.	Used for the canonical stopping-rule formulation and option-value interpretation.
Relevant past ASU macro quals in the local archive	2020 June: McCall search with many workers, asset market, layoffs, and stationary RCE. 2022 June: temporary unemployment compensation, benefit eligibility, quitting, and multiple reservation wages.	Used for Section 13, which gives exam-ready Bellman systems and proof outlines.

Module goal

This module is designed for a qualifying exam where the grader rewards a clean recursive formulation before algebra. The emphasis is: *timing, state variables, multiple value functions, reservation-wage proofs, economic and graphical intuition, dynamics, and comparative statics*. When in doubt, write the clock, define the value at the exact decision node, and only then simplify.

Detailed Blueprint

- Part I. Core architecture of a search problem.** Define the timing, observed shocks, state variables, value measurement point, and acceptance object. Learn when to use an offer-stage value $v(w)$, a pre-offer unemployment value U , and an employed value $\hat{v}(w)$.
- Part II. Canonical McCall search.** Set up the Bellman equation, prove existence by contraction, prove the reservation wage rule, derive the reservation wage equation in primitive form, compute unemployment duration, and draw the reservation graph.
- Part III. Comparative statics and intuition.** Convert the Bellman equation into a scalar equation in w^* before differentiating. Interpret $w^* - b$ as the cost of another search period and the tail integral as the option value of search. Cover b , β , offer probabilities, first-order stochastic dominance, mean-preserving spreads, and layoff risk.
- Part IV. Multiple value functions.** Work through layoffs, stochastic benefits, state-dependent offer probabilities, temporary jobs, recall, temporary UI eligibility, quitting, and asset-market variants. The rule is always: the reservation wage equates the value of accepting to the value of rejecting at the same decision node.
- Part V. Dynamics and equilibrium.** Move from an individual stopping problem to unemployment duration and aggregate stock-flow dynamics. State the stationary recursive competitive equilibrium for a many-worker McCall model with assets.
- Part VI. Exam templates.** Translate Problem Set 3 and the relevant past quals into reusable templates: Bellman equations, cutoffs, rankings, proof skeletons, and intuition paragraphs.
- Part VII. Pitfalls checklist.** Avoid common errors: integrating before maximizing when offers are observed, using the wrong continuation value, assuming state-dependent arrival implies state-dependent reservation wages, and differentiating a Bellman equation while ignoring that v is endogenous.

1 The Architecture of a Search Problem

Search problems are dynamic stopping problems. An unemployed worker receives opportunities over time, observes some payoff-relevant information, and chooses whether to stop searching by accepting an offer. The hard part on exams is rarely the algebra. The hard part is writing the value functions at the correct decision nodes.

1.1 The first line of every solution: the clock

Before writing a Bellman equation, record the event order. A typical clock is

state realized $\rightarrow$ current payoff or benefit $\rightarrow$ offer arrives $\rightarrow$ accept/reject $\rightarrow$ next state.

Small timing changes alter the Bellman equation. For example, if an accepted job starts next period, the accept value is often $\beta w/(1 - \beta)$ plus any current-period unemployment benefit. If the job starts immediately, the accept value is $w/(1 - \beta)$. The reservation wage can be the same or different depending on this convention, but the Bellman equation must match the problem statement.

Recipe: Bellman setup checklist

For any search question, fill in the following six objects.

1. **Individual state:** current employment status, current wage if employed, assets, UI eligibility, aggregate state, previous offer if recall is possible.
2. **Shock timing:** is the wage offer observed before the decision? is the aggregate state observed? is the separation shock before or after work?
3. **Current payoff:** wage w , benefit b , search cost $C(a)$, asset income, or zero.
4. **Continuation state:** unemployed next period, employed at wage w , eligible or ineligible for benefits, same aggregate state or Markov transition.
5. **Decision node:** place the max exactly where the worker chooses.
6. **Reservation object:** solve the indifference condition *accept value* = *reject value*.

1.2 Three common value measurements

The same model can be written in several equivalent ways. Confusion often comes from mixing them.

Formulation 1. Offer-stage value $v(w)$. Measure the value after the worker observes a wage offer w :

$$v(w) = \max\{\text{accept value at } w, \text{reject value}\}.$$

This is best when proving the reservation wage rule because the accept value is increasing in w while the reject value is constant in w .

Formulation 2. Unemployed pre-offer value U . Measure the value at the beginning of an unemployment period before the wage draw:

$$U = \text{flow payoff} + \beta \mathbb{E}[\max\{\text{accept continuation}, U\}].$$

This is compact and useful for multiple states.

Formulation 3. Employed value $\hat{v}(w)$. When jobs can end or workers can quit, the value of being employed at wage w is not simply $w/(1 - \beta)$:

$$\hat{v}(w) = w + \beta[(1 - \lambda)\hat{v}(w) + \lambda U]$$

if a job separates with probability λ .

Pitfall: Using the wrong value measurement

Do not write $v(w) = \max\{w/(1 - \beta), U\}$ unless $v(w)$ and U are measured at compatible dates. In the canonical model, if U is a pre-offer value and $v(w)$ is an offer-stage value, the exact relation is usually $U = b + \beta\mathbb{E}[v(W')]$, while the reservation condition is $w^*/(1 - \beta) = U$ only under the corresponding timing convention. State the timing and then the equality is obvious.

2 Canonical McCall Search

2.1 Environment

A risk-neutral worker maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t c_t, \quad \beta \in (0, 1).$$

While unemployed, the worker receives an offer $W \sim F$ on $[0, \bar{w}]$, independently over time. If she accepts wage w , she works forever at w . If she rejects, she receives unemployment benefit b today and searches again tomorrow. Assume $b < \bar{w}$ and no assets, no recall, and no on-the-job search.

Use the offer-stage value $v(w)$, measured after observing w . The Bellman equation is

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, b + \beta \int_0^{\bar{w}} v(w') dF(w') \right\}. \quad (1)$$

The accept value is $w/(1 - \beta)$. The reject value is constant in w because rejecting returns the worker to the same unemployment problem next period.

2.2 Contraction and existence

Let $\mathcal{C}$ be the set of bounded continuous functions on $[0, \bar{w}]$ with the sup norm. Define

$$(Tv)(w) = \max \left\{ \frac{w}{1 - \beta}, b + \beta \int v(w') dF(w') \right\}.$$

For any $v_1, v_2 \in \mathcal{C}$,

$$\begin{aligned} |(Tv_1)(w) - (Tv_2)(w)| &\leq \left| \beta \int (v_1(w') - v_2(w')) dF(w') \right| \\ &\leq \beta \|v_1 - v_2\|_{\infty}. \end{aligned}$$

Thus T is a contraction with modulus β and has a unique fixed point.

Exam-facing point: What to prove for a reservation wage

A reservation wage proof has three lines:

1. The accept value $A(w)$ is strictly increasing in w .
2. The reject value R is independent of w .
3. Therefore the acceptance set $\{w : A(w) \geq R\}$ is an interval of the form $[w^*, \bar{w}]$, with w^* pinned down by $A(w^*) = R$ if the crossing is interior.

Tie-breaking at $w = w^*$ does not affect the value. With atoms, state whether acceptance is weak or strict at the cutoff.

2.3 Reservation wage

Define the constant reject value

$$R \equiv b + \beta \int_0^{\bar{w}} v(w') dF(w')$$

and define w^* by

$$\frac{w^*}{1 - \beta} = R. \quad (2)$$

The optimal policy is

$$\sigma(w) = \begin{cases} \text{reject}, & w < w^*, \\ \text{accept}, & w \geq w^*. \end{cases}$$

Using (2), the value can be written as the upper envelope

$$v(w) = \frac{\max\{w, w^*\}}{1 - \beta}. \quad (3)$$

Substitute (3) into (2):

$$\begin{aligned} w^* &= (1 - \beta)b + \beta \int_0^{\bar{w}} \max\{w, w^*\} dF(w) \\ &= (1 - \beta)b + \beta w^* + \beta \int_{w^*}^{\bar{w}} (w - w^*) dF(w). \end{aligned} \quad (4)$$

Therefore

$$w^* = b + \frac{\beta}{1 - \beta} \int_{w^*}^{\bar{w}} (w - w^*) dF(w). \quad (5)$$

Define the tail option value

$$\mathcal{S}(x) \equiv \int_x^{\bar{w}} (w - x) dF(w) = \mathbb{E}[(W - x)^+]. \quad (6)$$

When integration by parts is valid,

$$\mathcal{S}(x) = \int_x^{\bar{w}} (1 - F(z)) dz. \quad (7)$$

Thus the reservation equation is

$$w^* - b = \frac{\beta}{1 - \beta} \mathcal{S}(w^*). \quad (8)$$

Intuition: Why $w^* > b$

If the worker rejects wage w^* , she keeps the option to draw future wages above w^* . The right side of (8) is exactly the discounted expected surplus from those future draws. Hence the reservation wage exceeds the unemployment benefit unless there is no chance of drawing a better wage.

2.4 Graphical representation

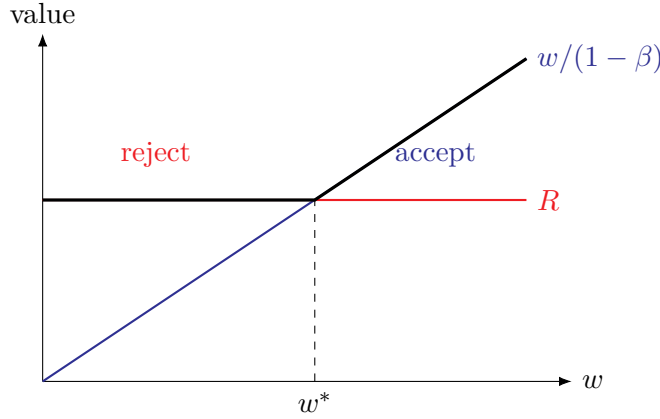


Figure 1: Reservation wage as the crossing of an increasing accept value and a constant reject value.

2.5 Unemployment duration

Let

$$p \equiv \Pr(W \geq w^*) = 1 - F(w^*)$$

when F has no atom at w^* . Then duration D is geometric:

$$\Pr(D = n) = (1 - p)^{n-1}p, \quad \mathbb{E}[D] = \frac{1}{p} = \frac{1}{1 - F(w^*)}.$$

A higher reservation wage reduces the acceptance probability and lengthens expected unemployment duration, holding F fixed.

Pitfall: Duration is endogenous

Do not say “higher benefits mechanically raise duration” without the intermediate step. Benefits raise duration because they raise w^* , and a higher w^* lowers $1 - F(w^*)$.

3 Comparative Statics in the Basic Model

The exam-safe method is to reduce the Bellman equation to the scalar reservation equation. Let

$$G(x; b, \beta, F) = x - b - \frac{\beta}{1 - \beta} \mathcal{S}(x; F), \quad G(w^*; b, \beta, F) = 0. \quad (9)$$

For differentiable F without atoms at x ,

$$\mathcal{S}_x(x) = -(1 - F(x)), \quad G_x(x) = 1 + \frac{\beta}{1 - \beta} (1 - F(x)) > 0. \quad (10)$$

This strict positivity gives uniqueness and signs comparative statics.

3.1 Unemployment benefit b

Differentiate $G(w^*; b, \beta, F) = 0$:

$$\frac{\partial w^*}{\partial b} = \frac{1}{G_x(w^*)} = \frac{1}{1 + \frac{\beta}{1-\beta}(1 - F(w^*))} > 0. \quad (11)$$

The derivative is less than or equal to one. A higher benefit makes waiting less costly, so the worker becomes more selective.

3.2 Discount factor β

Since

$$\frac{\partial}{\partial \beta} \left(\frac{\beta}{1-\beta} \right) = \frac{1}{(1-\beta)^2},$$

we get

$$\frac{\partial w^*}{\partial \beta} = \frac{\mathcal{S}(w^*)/(1-\beta)^2}{G_x(w^*)} > 0. \quad (12)$$

A more patient worker values future draws more and raises the reservation wage.

3.3 Offer arrival probability

Suppose an offer arrives with probability $\alpha \in (0, 1]$ each period, and no offer is equivalent to forced rejection. With the same timing as the U formulation below, the reservation equation becomes

$$w^* = b + \frac{\alpha\beta}{1-\beta}\mathcal{S}(w^*). \quad (13)$$

Then

$$\frac{\partial w^*}{\partial \alpha} = \frac{\frac{\beta}{1-\beta}\mathcal{S}(w^*)}{1 + \frac{\alpha\beta}{1-\beta}(1 - F(w^*))} > 0. \quad (14)$$

Better offer arrival raises the option value of remaining unemployed.

Exam-facing point: State-dependent offer arrival does not automatically imply state-dependent cutoffs

If the current state only affects the probability that an offer arrives, then conditional on already having an offer the current state may be irrelevant. In that case there can be a single reservation wage even though the value of being unemployed before the offer differs across states.

3.4 Wage distribution

If F_2 first-order stochastically dominates F_1 , then $F_2(z) \leq F_1(z)$ for all z , so

$$\mathcal{S}(x; F_2) = \int_x^{\bar{w}} (1 - F_2(z))dz \geq \int_x^{\bar{w}} (1 - F_1(z))dz = \mathcal{S}(x; F_1).$$

The right side of the reservation equation shifts up and w^* rises.

For a mean-preserving spread, the relevant object is the upper-tail integral $\int_x^{\bar{w}} (1 - F(z))dz$. If a higher-spread distribution increases this integral for every x , then $\mathcal{S}(x)$ rises and w^* rises. The intuition is that downside risk is protected by rejection, while upside risk raises the option value of search.

Pitfall: Comparative statics directly from the Bellman equation

A common bad answer says: “ b rises, so the RHS of the Bellman equation rises, so w^* rises.” The value function inside the integral also changes. The safe answer first derives (5), which depends only on primitives and w^* .

3.5 The U formulation

Let U be the value at the beginning of an unemployment period before the offer. Then

$$U = b + \beta \int_0^{\bar{w}} \max \left\{ \frac{w}{1-\beta}, U \right\} dF(w). \quad (15)$$

The reservation wage satisfies

$$\frac{w^*}{1-\beta} = U. \quad (16)$$

Substituting (16) into (15) yields the same scalar equation (5). This formulation is usually easier when the problem has aggregate states or no-offer probabilities.

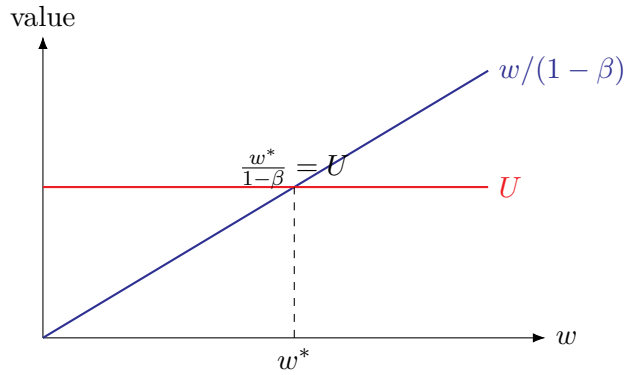


Figure 2: Alternative formulation: U is a horizontal continuation value, and the cutoff solves $w^*/(1-\beta) = U$.

4 Jobs That Can End: Layoffs and Stock-Flow Dynamics

Suppose an employed worker loses her job with probability $\lambda \in (0, 1)$ at the beginning or end of each period, depending on the problem clock. Under the common timing in which an employed worker at wage w receives wage w this period and next period is unemployed with probability λ , define

$$\hat{v}(w) = w + \beta[\lambda U + (1-\lambda)\hat{v}(w)], \quad (17)$$

$$U = b + \beta \int_0^{\bar{w}} \max\{\hat{v}(w), U\} dF(w). \quad (18)$$

Solving (17),

$$\hat{v}(w) = \frac{w + \beta\lambda U}{1 - \beta(1-\lambda)}. \quad (19)$$

This is strictly increasing in w , so the optimal policy is a reservation wage rule. At the cutoff, $\hat{v}(w^*) = U$, so

$$U = \frac{w^*}{1 - \beta}. \quad (20)$$

Substituting back gives

$$w^* - b = \frac{\beta}{1 - \beta(1 - \lambda)} \int_{w^*}^{\bar{w}} (w - w^*) dF(w). \quad (21)$$

When $\lambda = 0$, this collapses to the canonical McCall equation. As λ rises, the coefficient $\beta/[1 - \beta(1 - \lambda)]$ falls, so w^* falls. A high wage is less valuable when the job is less durable.

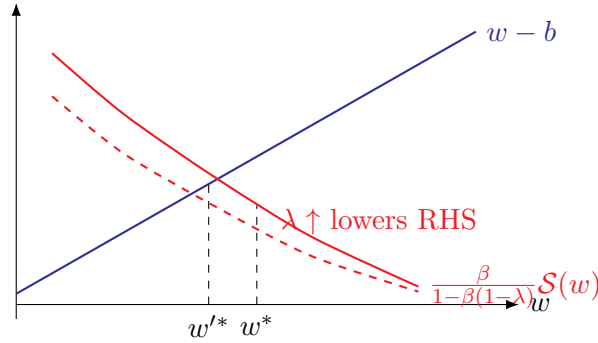


Figure 3: An increase in layoff risk lowers the option value of waiting for a high wage and reduces the reservation wage.

4.1 Aggregate unemployment dynamics

Let $p = 1 - F(w^*)$ be the job-finding probability among unemployed workers, and let u_t be the unemployment rate. With exogenous separation probability λ ,

$$u_{t+1} = (1 - p)u_t + \lambda(1 - u_t) = \lambda + (1 - p - \lambda)u_t. \quad (22)$$

The steady state is

$$u^* = \frac{\lambda}{p + \lambda}. \quad (23)$$

Convergence is monotone if $p + \lambda < 1$, oscillatory if $p + \lambda > 1$, and immediate if $p + \lambda = 1$.

Intuition: Two margins behind unemployment

Steady-state unemployment is high when separations are frequent (λ high) or when job finding is slow (p low). The individual reservation wage affects aggregate unemployment through $p = 1 - F(w^*)$.

5 Endogenous Search Intensity

Suppose the worker chooses search effort $a \in [0, 1]$. An offer arrives with probability a . Search effort has cost $C(a)$, with $C' > 0$, $C'' > 0$, $C(0) = 0$, $C'(0) = 0$, and $C'(a) \rightarrow \infty$ as $a \rightarrow 1$. There are no

layoffs. The unemployed value is

$$U = \max_{a \in [0,1]} \left\{ b - C(a) + \beta \left[(1-a)U + a \int_0^{\bar{w}} \max \left\{ \frac{w}{1-\beta}, U \right\} dF(w) \right] \right\}. \quad (24)$$

Let $w^* = (1-\beta)U$. For an interior optimum a^* , the reservation equation is

$$w^* = b - C(a^*) + \frac{a^*\beta}{1-\beta} \mathcal{S}(w^*), \quad (25)$$

and the effort FOC is

$$C'(a^*) = \frac{\beta}{1-\beta} \mathcal{S}(w^*). \quad (26)$$

Combining (25) and (26),

$$w^* = b - C(a^*) + a^* C'(a^*). \quad (27)$$

Equation (26) is downward sloping in (w^*, a^*) : a higher reservation wage lowers the probability-weighted gain from effort. Equation (27) is upward sloping because

$$\frac{d}{da} [-C(a) + aC'(a)] = aC''(a) > 0.$$

A higher benefit b shifts (27) up. The new equilibrium has higher w^* and lower a^* .

Exam-facing point: Moral hazard language

Unemployment insurance has two effects in this model: it raises selectivity, $w^* \uparrow$, and lowers search intensity, $a^* \downarrow$. The first effect lowers acceptance conditional on an offer; the second lowers the probability of receiving an offer.

6 State-Dependent Offer Arrival

This section covers the most common qualifying exam twist: aggregate states affect the probability of receiving an offer.

6.1 IID aggregate state and no-offer probability

Let the economy be in state $i \in \{\ell, h\}$, iid over time, with probability π_i . An unemployed worker receives an offer with probability p_i , where $0 < p_\ell < p_h < 1$. Conditional on receiving an offer, $W \sim F$ independently of i . Let

$$q \equiv \pi_\ell p_\ell + \pi_h p_h \quad (28)$$

be the unconditional offer-arrival probability next period.

Using a pre-offer value U ,

$$U = b + \beta \left[(1-q)U + q \int_0^{\bar{w}} \max \left\{ \frac{w}{1-\beta}, U \right\} dF(w) \right]. \quad (29)$$

The cutoff $w^* = (1-\beta)U$ solves

$$w^* = b + \frac{q\beta}{1-\beta} \mathcal{S}(w^*). \quad (30)$$

Because the current state is iid and affects only whether an offer arrives, the reservation wage conditional on having an offer is the same in both states. If π_h rises, then

$$\frac{dq}{d\pi_h} = p_h - p_\ell > 0,$$

so w^* rises by the arrival-probability comparative static.

Intuition: Why the cutoff is not state dependent here

Once the worker has an offer in hand, the current state no longer affects either the accepted job value or the reject continuation value. The state matters before the offer because it changes the chance that an offer arrives, but it does not matter at the accept/reject node.

7 Stochastic Unemployment Benefits

Now the aggregate state affects the benefit amount, not merely the arrival probability. Let $b_h = \bar{b}$ and $b_\ell = b$, with $b < \bar{b} < \bar{w}$. The high state persists with probability p and switches to low with probability $1 - p$; the low state is absorbing.

Use offer-stage values $V_h(w)$ and $V_\ell(w)$. Let

$$\bar{V}_s \equiv \int_0^{\bar{w}} V_s(w') dF(w'), \quad s \in \{h, \ell\}.$$

Then

$$V_\ell(w) = \max \left\{ \frac{w}{1 - \beta}, b + \beta \bar{V}_\ell \right\}, \quad (31)$$

$$V_h(w) = \max \left\{ \frac{w}{1 - \beta}, \bar{b} + \beta [p \bar{V}_h + (1 - p) \bar{V}_\ell] \right\}. \quad (32)$$

Define cutoffs w_ℓ^* and w_h^* by equality of accept and reject values:

$$\frac{w_\ell^*}{1 - \beta} = b + \beta \bar{V}_\ell, \quad (33)$$

$$\frac{w_h^*}{1 - \beta} = \bar{b} + \beta [p \bar{V}_h + (1 - p) \bar{V}_\ell]. \quad (34)$$

Since $V_s(w) = \max\{w, w_s^*\}/(1 - \beta)$,

$$w_\ell^* = (1 - \beta)b + \beta \mathbb{E} \max\{W, w_\ell^*\}, \quad (35)$$

$$w_h^* = (1 - \beta)\bar{b} + \beta [p \mathbb{E} \max\{W, w_h^*\} + (1 - p) \mathbb{E} \max\{W, w_\ell^*\}]. \quad (36)$$

Equivalently,

$$w_\ell^* = b + \frac{\beta}{1 - \beta} \mathcal{S}(w_\ell^*), \quad (37)$$

$$w_h^* = (1 - \beta)\bar{b} + \beta [p(w_h^* + \mathcal{S}(w_h^*)) + (1 - p)(w_\ell^* + \mathcal{S}(w_\ell^*))]. \quad (38)$$

Proposition 7.1. *If $\bar{b} > b$, then $w_h^* > w_\ell^*$. Moreover, an increase in the low benefit b raises w_ℓ^* and raises w_h^* .*

Proof. The low-state equation is the canonical reservation equation, so $dw_\ell^*/db > 0$. The high-state reject value is larger than the low-state reject value because it has a larger current benefit and a weakly better future benefit process; hence the high cutoff is higher.

For the formal cross-effect, write $M(x) = \mathbb{E} \max\{W, x\} = x + \mathcal{S}(x)$, so $M'(x) = F(x)$ at continuity points. Define

$$H(w_h, w_\ell) = w_h - (1 - \beta)\bar{b} - \beta[pM(w_h) + (1 - p)M(w_\ell)].$$

Then

$$H_{w_h} = 1 - \beta p F(w_h) > 0, \quad H_{w_\ell} = -\beta(1 - p)F(w_\ell) < 0.$$

Therefore

$$\frac{dw_h^*}{db} = \frac{\beta(1 - p)F(w_\ell^*)}{1 - \beta p F(w_h^*)} \frac{dw_\ell^*}{db} > 0.$$

□

Intuition: Why low-state benefits affect high-state cutoffs

A worker in the high state may become a low-state unemployed worker in the future. Raising b in the absorbing low state increases the continuation value after rejection in the high state, so it raises the high-state reservation wage even though the current high benefit $\bar{b}$ is unchanged.

8 Temporary Jobs

In a temporary-job model, accepting an offer does not end the problem forever. The worker works next period and then returns to unemployment. This changes the reservation wage because accepting a job makes the worker skip a future search opportunity.

8.1 General two-state formulation

Let $s \in \{h, \ell\}$ be the current aggregate state. An unemployed worker in state s receives an offer with probability π_s , with $0 < \pi_\ell < \pi_h < 1$. Jobs last one period. The worker receives b at the beginning of the current unemployment period. If she accepts wage w at date t , she works at w in date $t + 1$ and returns to unemployment at date $t + 2$.

Let U_s be the value at the beginning of an unemployment period after state s is realized. Let

$$\begin{aligned} A_s &\equiv \mathbb{E}[U_{s'} \mid s_t = s], & \text{one-step unemployed continuation,} \\ B_s &\equiv \mathbb{E}[U_{s''} \mid s_t = s], & \text{two-step unemployed continuation.} \end{aligned}$$

If no offer arrives or the offer is rejected, the continuation value is βA_s . If wage w is accepted, the continuation value is $\beta w + \beta^2 B_s$. Thus the state- s cutoff is

$$\beta w_s^* + \beta^2 B_s = \beta A_s \implies \boxed{w_s^* = A_s - \beta B_s.} \quad (39)$$

The Bellman equation is

$$U_s = b + (1 - \pi_s)\beta A_s + \pi_s \int_0^{\bar{w}} \max\{\beta A_s, \beta w + \beta^2 B_s\} dF(w). \quad (40)$$

Using (39),

$$U_s = b + \beta A_s + \pi_s \beta \int_{w_s^*}^{\bar{w}} (w - w_s^*) dF(w). \quad (41)$$

8.2 IID aggregate state

If the aggregate state is iid, then $A_s = B_s = \bar{U} \equiv pU_h + (1-p)U_\ell$ for both current states. Hence

$$w_h^* = w_\ell^* = (1-\beta)\bar{U}. \quad (42)$$

The beginning-of-period values differ because $\pi_h > \pi_\ell$, but the reservation wage conditional on an offer is the same.

Intuition: IID temporary jobs

Current state affects the chance of getting an offer today. Once an offer is in hand, accepting versus rejecting only changes future paths, and under iid states the future distribution is independent of today's state. Therefore the cutoff is common.

8.3 Persistent Markov aggregate state

Suppose the state follows a symmetric Markov chain with persistence $p > 1/2$:

$$P(h | h) = P(\ell | \ell) = p.$$

Conjecture $U_h > U_\ell$. Then

$$\begin{aligned} A_h - A_\ell &= (2p-1)(U_h - U_\ell) > 0, \\ B_h - B_\ell &= (2p-1)^2(U_h - U_\ell) > 0. \end{aligned}$$

Therefore, using (39),

$$w_h^* - w_\ell^* = (2p-1)[1 - \beta(2p-1)](U_h - U_\ell) > 0. \quad (43)$$

Intuition: Why the high-state cutoff is higher with persistence

In a persistent high state, rejecting today is attractive because tomorrow is likely to be high too, and offers arrive more often in high states. Accepting a temporary job makes the worker spend tomorrow working and unable to search. Thus the opportunity cost of accepting is higher in the high state.

Exam-facing point: Sketch to prove $U_h > U_\ell$

Use the Bellman operator defined by (40). Start value iteration from a constant vector $U_h^0 = U_\ell^0$. Because $\pi_h > \pi_\ell$, the first iterate satisfies $U_h^1 \geq U_\ell^1$. If $U_h \geq U_\ell$, persistence $p > 1/2$ implies $A_h \geq A_\ell$ and $B_h \geq B_\ell$, and the higher offer probability in h weakly raises the option value term. Hence the operator maps the cone $\{U_h \geq U_\ell\}$ into itself. Contraction gives a unique fixed point in that cone; strict inequality follows from $\pi_h > \pi_\ell$ and positive probability of acceptable offers.

9 Two Offers and Recall

This is a formulation-intensive problem. The worker receives b at the beginning of the unemployment period. Then she draws $w_1 \sim U[0, 1]$ and decides whether to accept it. If she rejects, she draws $w_2 \sim U[0, 1]$. Before deciding on w_2 , she learns whether w_1 is still available. Recall is possible with probability p .

Assume accepted jobs start next period. Let U be the value at the beginning of an unemployment period before the first draw, and define the final-stage wage-unit cutoff

$$x \equiv (1-\beta)U. \quad (44)$$

9.1 Final decision after rejecting the first offer

If recall is not possible, the worker accepts w_2 iff

$$\frac{w_2}{1-\beta} \geq U \iff w_2 \geq x.$$

If recall is possible, she chooses the best of w_1 , w_2 , and unemployment. She recalls w_1 iff

$$w_1 \geq \max\{w_2, x\}. \quad (45)$$

Thus recall can occur only when the first offer is good enough relative to the outside option and the second offer.

Define the wage-unit continuation value after rejecting first offer w_1 :

$$K(w_1; x) = p\mathbb{E}_{W_2} \max\{w_1, W_2, x\} + (1-p)\mathbb{E}_{W_2} \max\{W_2, x\}. \quad (46)$$

At the first decision node, current b and the factor β are common to accepting and rejecting, so the worker accepts w_1 iff

$$w_1 \geq K(w_1; x). \quad (47)$$

The first-stage cutoff $\bar{w}_1$ solves

$$\boxed{\bar{w}_1 = K(\bar{w}_1; x).} \quad (48)$$

For $W_2 \sim U[0, 1]$ and $a \geq x$,

$$\mathbb{E} \max\{W_2, x\} = x + \frac{(1-x)^2}{2}, \quad (49)$$

$$\mathbb{E} \max\{a, W_2, x\} = a + \frac{(1-a)^2}{2}. \quad (50)$$

Therefore, if $\bar{w}_1 \geq x$ and $p < 1$,

$$\bar{w}_1 = p \left[\bar{w}_1 + \frac{(1-\bar{w}_1)^2}{2} \right] + (1-p) \left[x + \frac{(1-x)^2}{2} \right]. \quad (51)$$

If $p = 1$, the worker never strictly wants to accept a first offer below the upper bound, because rejecting preserves the first offer for sure and adds a free second draw.

9.2 Ranking of cutoffs

At $w_1 = x$,

$$K(x; x) = \mathbb{E} \max\{W_2, x\} > x$$

whenever there is positive probability of drawing $W_2 > x$. Therefore the first-stage cutoff satisfies

$$\bar{w}_1 > x. \quad (52)$$

The worker is more selective on the first offer because rejecting buys another immediate draw and sometimes preserves the first offer.

9.3 Beginning-of-period value

The start value is

$$U = b + \beta \mathbb{E}_{W_1} \left[\max \left\{ \frac{W_1}{1 - \beta}, \frac{K(W_1; x)}{1 - \beta} \right\} \right], \quad x = (1 - \beta)U. \quad (53)$$

Equivalently, in wage units,

$$x = (1 - \beta)b + \beta \mathbb{E}_{W_1} \max\{W_1, K(W_1; x)\}. \quad (54)$$

Equations (46) and (54) characterize the solution.

Intuition: Why recall can happen here but not in basic McCall

In the basic one-offer-per-period model, an offer rejected last period was below the reservation wage. Next period the worker's outside option is the same, so the old rejected offer is still not worth accepting. In the two-offer model, rejecting the first offer can be a way to obtain a free second draw within the same period. The first offer may be above the final outside option x but below the first-stage cutoff $\bar{w}_1$. If the second offer is worse and recall is available, the worker recalls the first offer.

10 Temporary Unemployment Compensation and Quits

This section gives a reusable Bellman template for the 2022-style past qual. The key state variable is *benefit eligibility*. A worker receives unemployment compensation only during the first period of an unemployment spell. After that, she is unemployed without benefits until she accepts a job. An employed worker can quit; quitting requalifies her for one period of compensation, but she must work at least one period to requalify again.

The exact current-period payoff depends on the exam's timing. The following formulation matches the common interpretation in which accepting an offer starts employment next period, rejecting while eligible pays b today, and rejecting while ineligible pays zero today.

Let

- $v_e(w)$ be the value at the beginning of a period for a worker employed last period at wage w , before any quit decision.
- $v_1(w)$ be the value for an unemployed worker who has drawn offer w and is eligible for first-period unemployment compensation if she rejects.
- $v_+(w)$ be the value for an unemployed worker who has drawn offer w but is no longer eligible for compensation.

Let

$$\bar{v}_+ \equiv \int_0^{\bar{w}} v_+(w') dF(w'), \quad \bar{v}_1 \equiv \int_0^{\bar{w}} v_1(w') dF(w').$$

A useful Bellman system is

$$v_e(w) = \max \{w + \beta v_e(w), b + \beta \bar{v}_+\}, \quad (55)$$

$$v_1(w) = \max \{\beta v_e(w), b + \beta \bar{v}_+\}, \quad (56)$$

$$v_+(w) = \max \{\beta v_e(w), \beta \bar{v}_+\}. \quad (57)$$

If the problem specifies that a quitter draws immediately, replace the quit payoff in (55) with the appropriate current benefit plus the correct expected offer-stage value. The state variables and reservation logic are unchanged.

10.1 Reservation wage logic

The value $v_e(w)$ is increasing in w . Therefore all three decisions have cutoff rules:

- Employed worker continues iff $w \geq w_e^*$.
- Eligible unemployed worker accepts iff $w \geq w_1^*$.
- Ineligible unemployed worker accepts iff $w \geq w_+^*$.

The robust ranking is

$$w_1^* > w_+^*. \quad (58)$$

Eligibility raises the value of rejection, so newly unemployed eligible workers are more selective than longer-unemployed ineligible workers.

The quit threshold w_e^* is compared by equating the value of staying at the current wage to the value of quitting. A worker who accepted a lower wage is more likely to quit later. Since workers with longer prior unemployment spells are willing to accept lower wages, the model predicts they are weakly more likely to quit after employment.

Exam-facing point: Temporary UI proof structure

For this class of questions, do not try to solve all fixed points first. Prove monotonicity of $v_e(w)$, then use cutoff equalities. Ranking $w_1^* > w_+^*$ follows from the higher reject payoff for eligible unemployed workers. The quit prediction follows from selection into lower accepted wages among ineligible workers.

11 Many Workers, Assets, and Stationary RCE

A past qualifying exam variant embeds McCall search in a stationary recursive competitive equilibrium with a bond market and layoffs. The main goal is to define equilibrium cleanly.

11.1 Individual states

Let workers have assets $a \in A = [\underline{a}, \infty)$ or a bounded computational grid, and employment status $e \in \{U, E\}$. If employed, the individual state includes the current wage w . Let q be the price of a one-period bond paying one unit next period. Let ρ be the job-loss probability.

A standard timing is: current employment status and assets are known, the worker chooses savings, unemployed workers draw offers, accept/reject determines next period employment, and employed workers may separate. Because timing details vary, the following Bellman equations are a template rather than a unique convention.

Employed worker:

$$V^E(a, w) = \max_{a' \geq \underline{a}} \left\{ u(c_E) + \beta \left[(1 - \rho)V^E(a', w) + \rho V^U(a') \right] \right\}, \quad c_E = w + a - qa'. \quad (59)$$

Unemployed worker after observing offer w :

$$\tilde{V}^U(a, w) = \max \left\{ \underbrace{\max_{a' \geq \underline{a}} \{u(c_A) + \beta V^E(a', w)\}}_{\text{accept}}, \underbrace{\max_{a' \geq \underline{a}} \{u(c_R) + \beta V^U(a')\}}_{\text{reject}} \right\}, \quad (60)$$

where c_A and c_R must reflect the problem's current-period payoff convention. The pre-offer unemployed value is

$$V^U(a) = \int_0^{\bar{w}} \tilde{V}^U(a, w) dF(w). \quad (61)$$

If the problem states that saving occurs before the offer, introduce an intermediate value after the saving choice. Do not force the timing into (60); adapt the nodes.

11.2 Stationary recursive competitive equilibrium

A stationary RCE consists of

1. value functions $V^E, V^U, \tilde{V}^U$;
2. policy functions for savings $g^E(a, w), g^A(a, w), g^R(a, w)$ and an acceptance rule $\sigma(a, w) \in \{0, 1\}$;
3. a stationary distribution μ over individual states (E, a, w) and (U, a) ;
4. a bond price q ;

such that

- (i) Given q , the value functions solve the household Bellman equations and policies attain the maxima.
- (ii) The acceptance policy has reservation form when assumed or proven: accept iff $w \geq w^*(a)$.
- (iii) The distribution is stationary under the induced law of motion:

$$\mu' = \mathcal{T}(\mu; g^E, g^A, g^R, \sigma, F, \rho) = \mu.$$

This law moves rejected unemployed and separated employed workers into unemployment, accepted offers into employment at their accepted wage, and applies the savings policies to assets.

- (iv) Bond market clears. If the bond is in zero net supply,

$$\int a'(x) d\mu(x) = 0.$$

11.3 Why reservation wages can depend on assets

With concave utility and borrowing constraints, the value of unemployment insurance and the cost of waiting depend on assets. If an employed worker consumes more than an unemployed worker with the same asset level, the marginal utility of resources is lower when employed. A higher asset level relaxes the unemployed worker's liquidity pressure, making rejection less painful. This raises the reservation wage $w^*(a)$.

Exam-facing point: Asset-search monotonicity proof outline

To show $w^*(a)$ is increasing, define the net gain from accepting,

$$\Delta(a, w) = \text{accept value at } (a, w) - \text{reject value at } a.$$

The cutoff solves $\Delta(a, w^*(a)) = 0$. Show $\Delta_w > 0$. Then show $\Delta_a < 0$ under the condition that employed consumption exceeds unemployed consumption at the same asset level; by the envelope theorem this condition makes the marginal value of assets lower in employment than unemployment. The implicit function theorem gives

$$\frac{dw^*}{da} = -\frac{\Delta_a}{\Delta_w} > 0.$$

12 Problem Set 3 Templates

This section condenses the four assigned problem-set variants into exam-ready answer templates.

12.1 PS3.1: McCall search with a chance of no offer

Setup. State $i \in \{\ell, h\}$ is iid over time. Offer arrival probability is p_i , with $p_h > p_\ell$. Conditional wage distribution is F , independent of i .

Bellman. Let $q = \pi_h p_h + (1 - \pi_h) p_\ell$. A compact recursive formulation is

$$U = b + \beta \left[(1 - q)U + q \int \max \left\{ \frac{w}{1 - \beta}, U \right\} dF(w) \right].$$

The reservation wage satisfies

$$w^* = b + \frac{q\beta}{1 - \beta} \mathcal{S}(w^*).$$

Cutoff independence. The cutoff does not depend on the realized state because after an offer arrives, the current state affects neither the accepted value nor the rejection continuation value.

Comparative static. Since $dq/d\pi_h = p_h - p_\ell > 0$, a rise in π_h raises w^* .

12.2 PS3.2: Stochastic unemployment benefits

Setup. High benefit $\bar{b}$, low benefit b , low state absorbing, high persists with probability p .

Bellman and cutoffs. Use (31)–(36). The low-state cutoff solves the standard equation with benefit b . The high-state cutoff uses a continuation value that mixes high and low future unemployment values.

Comparisons.

$$w_h^* > w_\ell^*, \quad \frac{dw_\ell^*}{db} > 0, \quad \frac{dw_h^*}{db} > 0.$$

High benefits make rejection more attractive in high states. Low-state benefits matter for high-state decisions because the worker may transition to low.

12.3 PS3.3: Two consecutive offers with recall

State variable. After rejecting w_1 , the state includes w_1 because it may be recalled.

Final decision.

$$\text{No recall: accept } w_2 \iff w_2 \geq x = (1 - \beta)U.$$

$$\text{Recall possible: recall } w_1 \iff w_1 \geq \max\{w_2, x\}.$$

First decision. Define $K(w_1; x)$ by (46). Accept first offer iff $w_1 \geq K(w_1; x)$. The first cutoff $\bar{w}_1$ solves $\bar{w}_1 = K(\bar{w}_1; x)$ and satisfies $\bar{w}_1 > x$.

Recall intuition. Rejected first offers can be recalled because first-stage rejection buys a free option to sample w_2 .

12.4 PS3.4: Temporary jobs

Bellman. Use (40). The cutoff is $w_s^* = A_s - \beta B_s$.

IID state. Since future state distributions do not depend on current state,

$$w_h^* = w_\ell^*.$$

Persistent Markov state. If $p > 1/2$ and $U_h > U_\ell$,

$$w_h^* > w_\ell^*.$$

A high current state raises the opportunity cost of taking a one-period job because it makes tomorrow's search opportunity more valuable.

13 Past Qual Templates

13.1 2020 June: McCall search with assets and layoffs

What the problem tests. It combines incomplete markets, endogenous reservation wages, layoffs, stationary distributions, and market clearing. The central tasks are to define a stationary RCE and prove a monotonicity property of $w^*(a)$.

Answer skeleton.

1. Define individual state: unemployed asset a ; employed asset-wage pair (a, w) .
2. Write household Bellman equations with the bond price q and borrowing constraint $a' \geq \underline{a}$.
3. State the reservation rule $w \geq w^*(a)$.
4. Define policies and the stationary distribution μ over individual states.
5. Write the distribution law of motion and stationarity $\mu = \mathcal{T}(\mu)$.
6. Impose bond market clearing.
7. For $w^*(a)$ increasing, use the net-gain function $\Delta(a, w)$ and the envelope theorem as in Section 11.

13.2 2022 June: Temporary unemployment compensation

What the problem tests. It asks whether you can track multiple unemployment states and compare reservation wages. The state is not just employed/unemployed; it includes eligibility for benefits and current wage if employed.

Answer skeleton.

1. Define $v_e(w)$, $v_1(w)$, and $v_+(w)$ exactly as in the prompt.
2. Write the Bellman system. Adjust current payoffs according to the prompt's timing.
3. Prove $v_e(w)$ is increasing in w .
4. Use monotonicity to prove cutoff rules w_e^*, w_1^*, w_+^* .
5. Rank unemployed cutoffs: $w_1^* > w_+^*$ because eligible rejection pays b and ineligible rejection does not.
6. Explain quits: longer prior unemployment spells are associated with lower accepted wages; lower accepted wages are closer to or below the quit threshold.

14 Graphical and Economic Intuition Bank

14.1 MC-MB diagram

The primitive reservation equation can be written

$$\underbrace{w^* - b}_{\text{marginal cost of rejecting } w^*} = \underbrace{\frac{\beta}{1-\beta} \mathcal{S}(w^*)}_{\text{marginal benefit / option value of search}}.$$

The left side is increasing in w^* . The right side is decreasing in w^* . Their unique intersection is the reservation wage.

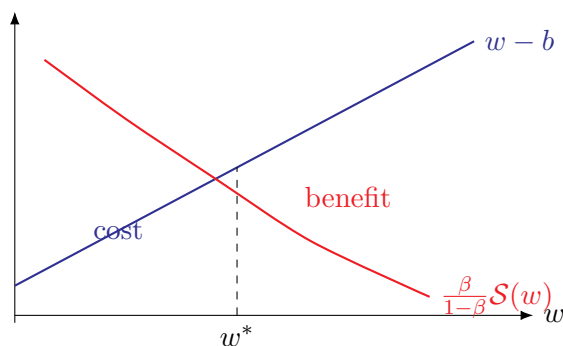


Figure 4: MC-MB graph for the reservation wage.

14.2 How shifts work

Change	Graph shift	Economic force
$b \uparrow$	$w - b$ shifts down	Waiting is less costly; $w^* \uparrow$.
$\beta \uparrow$	Option-value curve shifts up	Future offers matter more; $w^* \uparrow$.
Offer probability $\alpha \uparrow$	Option-value curve shifts up	Search is more productive; $w^* \uparrow$.
Mean-preserving spread	Option-value curve shifts up	Rejection insures downside; upside raises option value; $w^* \uparrow$.
Layoff risk $\lambda \uparrow$	Option-value curve shifts down	High accepted wages are less durable; $w^* \downarrow$.
Search cost $C(a) \uparrow$	Lowers net flow value and effort	Search intensity falls; cutoff effect depends on exact change, but higher UI gives $w^* \uparrow, a^* \downarrow$.

15 Common Pitfalls and Fast Fixes

Pitfall: Putting expectations outside the max

If the wage is observed before accepting, write $\int \max\{A(w), R\}dF(w)$, not $\max\{\int A(w)dF(w), R\}$. The latter describes a worker who must accept or reject before seeing the wage.

Pitfall: Forgetting the state that makes the problem recursive

Recall problems require previous offers as states. UI problems require eligibility status. Layoff or quit problems require employed wage. Asset problems require assets and the distribution for equilibrium.

Pitfall: Assuming all good states raise the cutoff

A good state raises the cutoff only if it raises the value of rejecting relative to accepting. If the state only affects offer arrival and the offer has already arrived, the cutoff may be state independent. If accepting a temporary job makes the worker skip a valuable future high-state search opportunity, the high state can raise the cutoff.

Pitfall: Ignoring current-period payoffs

Some prompts say an accepted job starts next period; others say employment begins immediately. This changes whether current b or current w enters the accept value. Draw the clock and keep current payoffs on both sides of the accept/reject comparison.

Pitfall: Differentiating values instead of primitive cutoff equations

For basic comparative statics, first derive a scalar equation in w^* . For multiple-state problems, use monotone operators or implicit systems. Do not treat $\int v(w)dF(w)$ as fixed when a primitive changes.

16 Formula Sheet

Model	Key equations
Basic McCall	$v(w) = \max\{w/(1 - \beta), b + \beta \mathbb{E}v(W')\}$; $w^* = b + \frac{\beta}{1-\beta} \mathcal{S}(w^*)$.
Expected duration	$p = 1 - F(w^*)$; $\mathbb{E}D = 1/p$.
Offer arrival α	$w^* = b + \frac{\alpha\beta}{1-\beta} \mathcal{S}(w^*)$.
Layoffs	$\hat{v}(w) = w + \beta[(1 - \lambda)\hat{v}(w) + \lambda U]$; $w^* - b = \frac{\beta}{1-\beta(1-\lambda)} \mathcal{S}(w^*)$.
Unemployment dynamics	$u_{t+1} = (1 - p)u_t + \lambda(1 - u_t)$; $u^* = \lambda/(p + \lambda)$.
Search intensity	$U = \max_a \{b - C(a) + \beta[(1 - a)U + a \mathbb{E} \max(W/(1 - \beta), U)]\}$; $C'(a^*) = \frac{\beta}{1-\beta} \mathcal{S}(w^*)$; $w^* = b - C(a^*) + a^* C'(a^*)$.
Stochastic benefits	$w_\ell^* = (1 - \beta)b + \beta \mathbb{E} \max\{W, w_\ell^*\}$; $w_h^* = (1 - \beta)\bar{b} + \beta[p \mathbb{E} \max\{W, w_h^*\} + (1 - p) \mathbb{E} \max\{W, w_\ell^*\}]$.
Temporary jobs	$w_s^* = A_s - \beta B_s$, where $A_s = \mathbb{E}[U_{t+1} s_t = s]$, $B_s = \mathbb{E}[U_{t+2} s_t = s]$.
Recall	Final cutoff $x = (1 - \beta)U$; recall w_1 iff $w_1 \geq \max\{w_2, x\}$; first cutoff solves $\bar{w}_1 = K(\bar{w}_1; x)$.

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